

Notes: Baqaee & Farhi (Econometrica, 2024)

Networks, Barriers, and Trade

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1 Big picture

- **Question.** How do productivity shocks, tariffs, trade costs, markups, nominal rigidities, and other wedge-like distortions affect real GDP and welfare in open economies with rich production networks?
- **Core contribution.** The paper builds a general analytical toolbox for large-scale trade models with international input–output linkages and arbitrary wedges.
- **Main idea.** Instead of solving a specific stylized model, the authors characterize changes in prices, quantities, real GDP, and welfare using sufficient statistics: input–output shares, Domar weights, elasticities of substitution, factor income shares, and wedges.
- **Two accounting questions.** The paper asks both ex post and ex ante questions: how to decompose observed growth, and how to predict counterfactual responses to shocks.
- **Real GDP versus welfare.** Real GDP and welfare are not the same object in open economies. Domestic real GDP is production-side output; welfare/real GNE depends on the prices of consumption baskets, transfers, and factoral terms-of-trade.
- **Real GDP decomposition.** To a first order, real GDP changes can be split into a direct technology effect and a reallocation effect. Reallocation effects vanish at an efficient initial allocation but matter when there are wedges.
- **Welfare decomposition.** Welfare changes also split into direct technology effects and reallocation effects, but welfare reallocation effects can be first order even without domestic distortions because countries buy and sell factors indirectly through trade.
- **Comparative statics.** The paper generalizes hat algebra to large open-economy production networks. A linear system determines all first-order changes in prices and sales shares.
- **Second-order losses.** For tariffs and other wedges around an efficient allocation, GDP losses are approximately Domar-weighted Harberger triangles.
- **Why networks matter.** Input–output linkages, low elasticities of substitution, sector-specific factors, and nominal rigidities magnify the welfare cost of trade barriers.
- **Quantitative takeaway.** In calibrated world-economy exercises, removing input–output linkages makes welfare losses too small; adding sticky wages can make trade disruptions much more costly.
- **Broad takeaway.** The paper bridges transparent theory and large-scale computation: it tells us which microeconomic objects drive the answer in quantitative trade counterfactuals.

2 Data, objects, and measurement

2.1 Why this is not a standard empirical paper

- The paper is mainly theoretical and quantitative, not a causal empirical design.
- The main objects are equilibrium statistics from an open-economy production network.
- The figures use stylized examples and quantitative exercises calibrated with world input–output data.

Interpretation. The paper’s empirical role is to discipline the sufficient statistics used by the theory, not to estimate a treatment effect.

2.2 Economy and agents

- There are many countries, households, producers, goods, and factors.

- Producers buy intermediate inputs from domestic and foreign suppliers.
- Households consume final goods and own factors or receive transfers.
- Wedges can represent markups, tariffs, iceberg costs, sticky prices, sticky wages, taxes, or other distortions.

Interpretation. The framework is deliberately broad. The same accounting formulas can be used for many trade models, including Armington-style models, nested-CES production networks, and models with nominal rigidities.

2.3 Input–output shares and Domar weights

Let Ω be the revenue-based input–output matrix and $\Psi = (I - \Omega)^{-1}$ the associated Leontief inverse. Let λ_i be the Domar weight or sales share of good i relative to the relevant aggregate. The cost-based versions are denoted by tildes, for example $\tilde{\Omega}$ and $\tilde{\Psi}$.

- Revenue-based shares describe who buys from whom at observed revenues.
- Cost-based shares adjust for wedges, so they describe marginal cost exposure.
- Domar weights measure the aggregate importance of a producer or input, including indirect exposure through supply chains.

Interpretation. The distinction between revenue-based and cost-based networks is essential when wedges such as markups or tariffs are present. A wedge means that revenue and marginal cost are not the same object.

2.4 Real GDP and welfare

- Real GDP of country c , denoted Y_c , is a production-side object.
- Welfare or real GNE of household/country c , denoted W_c , is a consumption-side object.
- In closed or frictionless environments, the two may coincide more closely.
- In open economies, they can differ because foreign productivity shocks and factor-price changes affect the purchasing power of domestic households even if domestic real output does not move.

Interpretation. The paper repeatedly emphasizes that real GDP and welfare have different sufficient statistics in open economies.

2.5 The quantitative input data

- The growth-accounting exercise uses world input–output data to compare cumulative decompositions of real GNE for the United States and Italy.
- The counterfactual exercise calibrates a multicountry, multisector world economy using input–output shares and nested-CES elasticities.
- The calibrated model compares benchmark losses from higher iceberg trade costs to losses under alternative assumptions: no input–output linkages, no complementarities, no factor mobility, and sticky wages.

Interpretation. The same microeconomic statistics used in the theorems are also the inputs for the quantitative counterfactuals.

3 Models and theoretical specifications

3.1 General decomposition logic

For any aggregate object W_c , the paper starts from the identity

$$d \log W_c = \frac{\partial \log W_c}{\partial \log A} d \log A + \frac{\partial \log W_c}{\partial \log L} d \log L + \frac{\partial \log W_c}{\partial \mathcal{X}} d \mathcal{X}. \quad (1)$$

The first two terms are changes in technologies and factor supplies. The last term is the change in the allocation matrix.

Interpretation. The goal of the paper is to express these terms in measurable network statistics and then use them for growth accounting and counterfactuals.

3.2 Theorem 1: real GDP

The first central result decomposes country real GDP:

$$d \log Y_c = \underbrace{\sum_{i \in N_c} \tilde{\lambda}_i^{Y_c} d \log A_i + \sum_{f \in F_c} \tilde{\Lambda}_f^{Y_c} d \log L_f + \sum_{k \in N - N_c} (\tilde{\Lambda}_k^{Y_c} - \Lambda_k^{Y_c}) d \log q_{ck}}_{\text{technology and imported-input effects}} + \underbrace{\text{wedges and changing factor/material shares}}_{\text{reallocation terms}}. \quad (2)$$

Interpretation. Real GDP responds directly to domestic productivity, domestic factor supply, and imported intermediate inputs. It also responds to reallocations when the initial allocation is distorted.

domestic producers holding fixed domestic productivity, factors, and imported materials).

THEOREM 1—Real GDP: *The change in real GDP of country c in response to productivity shocks, factor supply shocks, transfer shocks, and shocks to wedges is to a first order,¹⁴*

$$\begin{aligned} d \log Y_c = & \underbrace{\sum_{i \in N_c} \tilde{\lambda}_i^{Y_c} d \log A_i + \sum_{f \in F_c} \tilde{\Lambda}_f^{Y_c} d \log L_f + \sum_{k \in N - N_c} (\tilde{\Lambda}_k^{Y_c} - \Lambda_k^{Y_c}) d \log(q_{ck})}_{\Delta \text{ technology}} \\ & - \underbrace{\sum_{i \in N_c} \tilde{\lambda}_i^{Y_c} d \log \mu_i - \sum_{f \in F_c} \tilde{\Lambda}_f^{Y_c} d \log \Lambda_f^{Y_c} + \sum_{k \in N - N_c} (\Lambda_k^{Y_c} - \tilde{\Lambda}_k^{Y_c}) d \log \Lambda_k^{Y_c}}_{\Delta \text{ allocation}}. \quad (3) \end{aligned}$$

¹⁴Since there may be multiple equilibria, technically, $\mathcal{X}(A, L, \mu, T)$ is a correspondence. In this case, we

Figure 1: Original-paper excerpt: Theorem 1, real GDP decomposition.

3.3 Corollary: efficient allocation and Hulten's theorem

If the initial allocation is efficient, the reallocation terms in Theorem 1 vanish, so

$$d \log Y_c = \sum_{i \in N_c} \lambda_i^{Y_c} d \log A_i + \sum_{f \in F_c} \Lambda_f^{Y_c} d \log L_f. \quad (3)$$

Interpretation. This is the open-economy analogue of Hulten’s theorem for real GDP: at an efficient allocation, first-order GDP changes are Domar-weighted direct shocks.

3.4 Theorem 2: welfare

The welfare decomposition is

$$d \log W_c = \underbrace{\sum_{i \in N} \tilde{\lambda}_i^{W_c} d \log A_i + \sum_{f \in F} \tilde{\Lambda}_f^{W_c} d \log L_f}_{\text{technology}} + \underbrace{\left[- \sum_{i \in N} \tilde{\lambda}_i^{W_c} d \log \mu_i + \sum_{f \in F^*} (\Lambda_f^c - \tilde{\Lambda}_f^{W_c}) d \log \Lambda_f + \frac{dT_c}{\text{GNE}_c} \right]}_{\text{allocation and terms-of-trade effects}}. \quad (4)$$

Interpretation. Welfare depends on foreign productivity through the cost of the consumption basket. It also depends on factorial terms-of-trade: whether the household is a net seller or buyer of factors whose prices change.

Figure 2: We now turn our attention to changes in welfare (real GNE).

THEOREM 2—Welfare: *The change in welfare of country c in response to productivity shocks, factor supply shocks, and transfer shocks can be written as*

$$\begin{aligned} d \log W_c = & \underbrace{\sum_{i \in N} \tilde{\lambda}_i^{W_c} d \log A_i + \sum_{f \in F} \tilde{\Lambda}_f^{W_c} d \log L_f}_{\Delta \text{ technology}} \\ & - \underbrace{\sum_{i \in N} \tilde{\lambda}_i^{W_c} d \log \mu_i + \sum_{f \in F^*} (\Lambda_f^c - \tilde{\Lambda}_f^{W_c}) d \log \Lambda_f + \frac{dT_c}{\text{GNE}_c}}_{\Delta \text{ allocation}}, \end{aligned} \quad (5)$$

where $\tilde{\Lambda}_f^{W_c} = 0$ whenever f is a fictitious factor. The change $d \log W$ of world real GNE is obtained by suppressing the country index c . That is,

$$d \log W = \underbrace{\sum_{i \in N} \tilde{\lambda}_i^W d \log A_i + \sum_{f \in F} \tilde{\Lambda}_f^W d \log L_f}_{\Delta \text{ technology}} - \underbrace{\sum_{i \in N} \tilde{\lambda}_i^W d \log \mu_i - \sum_{f \in F} \tilde{\Lambda}_f^W d \log \Lambda_f}_{\Delta \text{ allocation}}.$$

Figure 2: Original-paper excerpt: Theorem 2, welfare decomposition.

3.5 Real GDP versus welfare

- A foreign productivity improvement may leave domestic real GDP unchanged if domestic technology and factors do not change.
- The same foreign productivity improvement can raise domestic welfare because imported goods and inputs become cheaper.
- This distinction is why the paper separates real GDP accounting from welfare accounting.

Interpretation. The difference is not a technicality: trade counterfactuals are often welfare questions, not only GDP questions.

3.6 Nested-CES standard form

For counterfactual comparative statics, the paper specializes to nested-CES aggregators written in standard form:

$$p_k = \frac{\mu_k}{A_k} \left(\sum_{j \in N+F_c} \tilde{\Omega}_{kj} p_j^{1-\theta_k} \right)^{1/(1-\theta_k)}. \quad (5)$$

Here θ_k is the elasticity of substitution for producer or consumption nest k .

Interpretation. The standard-form representation treats each CES aggregator as a separate producer, which makes the input–output network and elasticities sufficient for comparative statics.

3.7 Input–output covariance

The paper defines an input–output covariance operator:

$$\text{Cov}_{\tilde{\Omega}(k)}(\Psi_{(i)}, \Psi_{(j)}) = \sum_l \tilde{\Omega}_{kl} \Psi_{li} \Psi_{lj} - \left(\sum_l \tilde{\Omega}_{kl} \Psi_{li} \right) \left(\sum_l \tilde{\Omega}_{kl} \Psi_{lj} \right). \quad (6)$$

Interpretation. This covariance measures whether two inputs or shocks tend to travel through the same parts of the production network. It is the key statistic for substitution effects in network economies.

3.8 Theorem 3: prices and sales shares

Theorem 3 gives the first-order response of prices and sales shares. The price equation captures forward propagation:

$$d \log p_i = \sum_{k \in N} \tilde{\Psi}_{ik} (d \log \mu_k - d \log A_k) + \sum_{f \in F} \tilde{\Psi}_{if} d \log \Lambda_f. \quad (7)$$

The sales-share equation captures backward propagation:

$$d \log \lambda_i = \text{downstream wedge terms} + \text{substitution terms} + \text{redistribution terms}. \quad (8)$$

Interpretation. Shocks move downstream through prices and upstream through expenditure shares. Solving the resulting linear system gives all first-order equilibrium changes.

first order,

$$d \log p_i = \sum_{k \in N} \tilde{\Psi}_{ik} (d \log \mu_k - d \log A_k) + \sum_{f \in F} \tilde{\Psi}_{if} d \log \Lambda_f. \quad (7)$$

To a first order, the change in the sales share of a good or factor $i \in N + F$ is

$$\begin{aligned} d \log \lambda_i = & \sum_{k \in N+F} \left(\mathbf{1}_{\{i=k\}} - \frac{\lambda_k}{\lambda_i} \Psi_{ki} \right) d \log \mu_k + \sum_{k \in N} \frac{\lambda_k}{\lambda_i} \mu_k^{-1} (1 - \theta_k) \text{Cov}_{\tilde{\Omega}(k)}(\Psi_{(i)}, d \log p) \\ & + \sum_{g \in F^*} \sum_{c \in C} \frac{\lambda_i^{W_c} - \lambda_i}{\lambda_i} \Phi_{cg} \Lambda_g d \log \Lambda_g, \end{aligned} \quad (8)$$

where $d \log p$ is the $(N + F) \times 1$ vector of price changes in (7). The change in wedge income accruing to household c (represented by a fictitious factor) is

$$d \log \Lambda_c = \sum_i \frac{\Phi_{ci} \lambda_i}{\Lambda_c} (\mu_i^{-1} d \log \mu_i + (1 - \mu_i^{-1}) d \log \lambda_i). \quad (9)$$

Figure 3: Original-paper excerpt: Theorem 3, price and sales-share comparative statics.

3.9 Differential exact-hat algebra

- Traditional exact-hat algebra solves a nonlinear system once.
- The paper's differential method solves a linearized system repeatedly along a path.
- This provides exact nonlinear comparative statics when iterated and can be computationally faster in large nonlinear models.

Interpretation. The method preserves the transparency of first-order formulas while still allowing large counterfactual shocks.

3.10 Theorem 4: world welfare

At an efficient equilibrium, the second-order change in world welfare is

$$\Delta \log \delta \approx \Delta \log Y + \text{Cov}_{\chi^w}(\Delta \log \chi_c^W, \Delta \log P_{W_c}). \quad (9)$$

Interpretation. World welfare equals world real GDP plus a redistributive covariance term. If income shifts toward households facing higher consumer price increases, the covariance term matters.

$$W^{\text{BS}}(\delta\bar{W}_1, \dots, \delta\bar{W}_c) = W^{\text{BS}}(W_1, \dots, W_c),$$

where $\bar{W}_c$ and W_c are the values at the initial and final equilibrium.

THEOREM 4—World Welfare: *Starting at an efficient equilibrium in response to change in wedges or technologies, changes in world welfare are given up to the second order by*

$$\Delta \log \delta \approx \Delta \log Y + \text{Cov}_{\Omega_{\chi^W}}(\Delta \log \chi_c^W, \Delta \log P_{W_c}).$$

Here, $\Delta \log \chi_c^W$ and $\Delta \log P_{W_c}$ are the change in country c 's nominal GNE and consumer price index, respectively.

Figure 4: Original-paper excerpt: Theorem 4, world welfare.

3.11 Second-order technology and wedge effects

For technology shocks, the second-order response of real GDP includes changes in Domar weights:

$$\Delta \log Y_c \approx \sum_i \lambda_i^{Y_c} \Delta \log A_i + \sum_f \Lambda_f^{Y_c} \Delta \log L_f + \frac{1}{2} \sum_i \Delta \log \lambda_i^{Y_c} \Delta \log A_i + \frac{1}{2} \sum_f \Delta \log \Lambda_f^{Y_c} \Delta \log L_f. \quad (10)$$

Interpretation. Microeconomic details matter for second-order productivity effects only through how they change sales shares and factor shares.

From an efficient equilibrium, the effect of nonlinearities on real GDP. Therefore, we have the following.

COROLLARY 4—Real GDP Response to Technology Shocks: *In the absence of wedges, the response of real GDP for each country c to productivity, factor endowment, and wedge shocks is, to a second-order approximation,*

$$\Delta \log Y_c \approx \sum_{i \in N_c} \lambda_i^{Y_c} \Delta \log A_i + \sum_{f \in F_c} \Lambda_f^{Y_c} \Delta \log L_f + \frac{1}{2} \sum_{i \in N_c} \Delta \lambda_i^{Y_c} \Delta \log A_i + \frac{1}{2} \sum_{f \in F_c} \Delta \Lambda_f^{Y_c} \Delta \log L_f.$$

For world GDP, suppress the country subscript c .

Figure 5: Original-paper excerpt: Corollary 4, second-order productivity effects.

3.12 Theorems 5 and 6: tariffs, wedges, and Harberger triangles

Starting from an efficient equilibrium, small tariffs or wedge-like distortions reduce real GDP approximately by Domar-weighted deadweight-loss triangles:

$$\Delta \log Y_c \approx \frac{1}{2} \sum_{i \in N_c} \lambda_i^{Y_c} \Delta \log y_i \Delta \log \mu_i. \quad (11)$$

Theorem 6 expresses these triangles in microeconomic primitives: wedges, elasticities, input–output covariances, and changes in factor prices.

Interpretation. Tariff losses are not only about direct imports. Networks and complementarities can magnify losses by changing how quantities move throughout the production system.

weighted sum of deadweight-loss triangles. We then express these deadweight-loss triangles in terms of microeconomic primitives.

THEOREM 5—Real GDP: *Starting at an efficient equilibrium, up to the second order, in response to the introduction of small tariffs or other distortions, changes in the real GDP of country c are given by*

$$\Delta \log Y_c \approx \frac{1}{2} \sum_{i \in N_c} \lambda_i^Y \Delta \log y_i \Delta \log \mu_i.$$

Changes in world real GDP (and real GNE) are given by suppressing the country subscript.

(a) Theorem 5: Domar-weighted Harberger triangles.

Corollary 3.³⁰

THEOREM 6—Real GDP: *Starting at an equilibrium without distortions, in response to the introduction of small tariffs or other distortions, the change in real GDP of country c is*

$$\begin{aligned} \Delta \log Y_c \approx & -\frac{1}{2} \sum_{i \in N_c} \sum_{k \in N} \Delta \log \mu_k \Delta \log \mu_i \sum_{j \in N} \lambda_j^Y \theta_j \text{Cov}_{\Omega(i)}(\Psi_{(k)}, \Psi_{(i)}) \\ & -\frac{1}{2} \sum_{i \in N_c} \sum_{g \in F} \Delta \log \Lambda_g \Delta \log \mu_i \sum_{j \in N} \lambda_j^Y \theta_j \text{Cov}_{\Omega(i)}(\Psi_{(g)}, \Psi_{(i)}) \\ & +\frac{1}{2} \sum_{i \in N_c} \sum_{c \in C} \chi_c^w \Delta \log \chi_c^w \Delta \log \mu_i (\lambda_i^w - \lambda_i) / \chi_c^Y. \end{aligned}$$

(b) Theorem 6: primitive network expression.

Figure 6: Original-paper excerpts: second-order tariff and wedge effects.

4 Design, interpretation, and what makes the results useful

4.1 Identification in a theoretical paper

- There is no treatment-control identification design.
- Credibility comes from general equilibrium accounting identities, envelope arguments, and comparative static derivations.
- Quantitative exercises illustrate the theoretical sufficient statistics using calibrated input–output data.

Interpretation. The paper’s identification is mathematical rather than econometric: given the model class and the sufficient statistics, the formulas characterize the equilibrium response.

4.2 Why details matter

- Input–output linkages increase exposure because goods are used indirectly through supply chains.
- Complementarities magnify costs because it is hard to substitute away from disrupted or taxed goods.
- Sector-specific factors magnify losses because resources cannot easily move to sectors where they are more valuable.
- Sticky wages can transform trade disruptions into unemployment and much larger welfare costs.

Interpretation. Large-scale models matter because these features are hard to summarize with a single trade share or a single elasticity.

4.3 Why Domar weights are central

- A producer can matter more than its direct value added because its output is used repeatedly as an intermediate input.
- Domar weights capture these direct and indirect exposures.
- Changes in Domar weights are the key second-order terms in productivity counterfactuals.

Interpretation. Domar weights are the bridge from micro shocks to aggregate outcomes.

4.4 Limits and assumptions

- The most explicit comparative statics use nested-CES structures, even though the broader framework is more general.
- Quantitative predictions require calibrated elasticities of substitution, which can be uncertain.
- The model is powerful for comparative statics but does not itself estimate the political or institutional causes of tariffs and trade barriers.
- The exact numerical losses in counterfactuals depend on the chosen baseline input–output data and elasticities.

Interpretation. The paper is best read as a general analytical toolkit rather than as a single empirical estimate.

5 Figures and original-paper excerpts

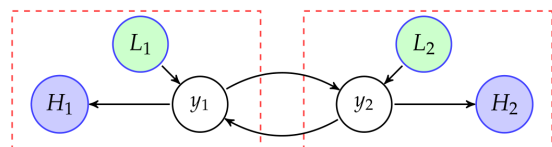
This section follows the compact notes format. Adjacent visuals are grouped to save space and improve comparison. The paper has no empirical tables; the visuals are the original figures plus the central theorem excerpts cropped from the paper.

5.1 Figure 1 and Figure 2: networks and nonlinear welfare losses

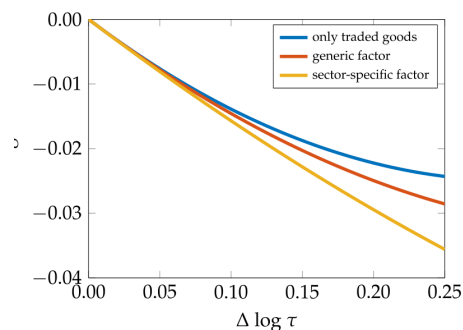
Read:

- Figure 1 shows the simplest two-country production network: each country has a factor, a producer, and a household, and goods cross borders through intermediate-input flows.
- Figure 2 compares welfare losses from increasing iceberg trade costs under three models that are first-order equivalent.
- With only traded goods, losses are modestly nonlinear.
- With generic factors and nontraded sectors, losses are larger.
- With sector-specific factors, losses are largest because resources cannot easily move across sectors.

Economic message. Even if models agree to first order, their second-order welfare losses can differ greatly. Complementarities and factor immobility make trade barriers more costly.



(a) Figure 1: two-country network.



(b) Figure 2: welfare losses from iceberg costs.

Figure 7: Original figures: input–output linkages and nonlinear losses.

5.2 Figure 3: decomposition of real GNE growth

Read:

- The figure compares two decompositions of cumulative real GNE growth for the United States and Italy.
- The blue line is real GNE, and the red line is real GDP/technology contribution.
- The yellow line is the terms-of-trade or factoral terms-of-trade component.
- The purple line is transfers.
- For the United States, real GNE growth closely tracks real GDP growth, with transfers and terms-of-trade playing smaller roles.
- For Italy, factoral terms-of-trade dynamics are more important and become negative after the late 2000s.

Economic message. Welfare changes can be decomposed into production, terms-of-trade, and transfer components. Open-economy welfare accounting is richer than domestic GDP accounting.

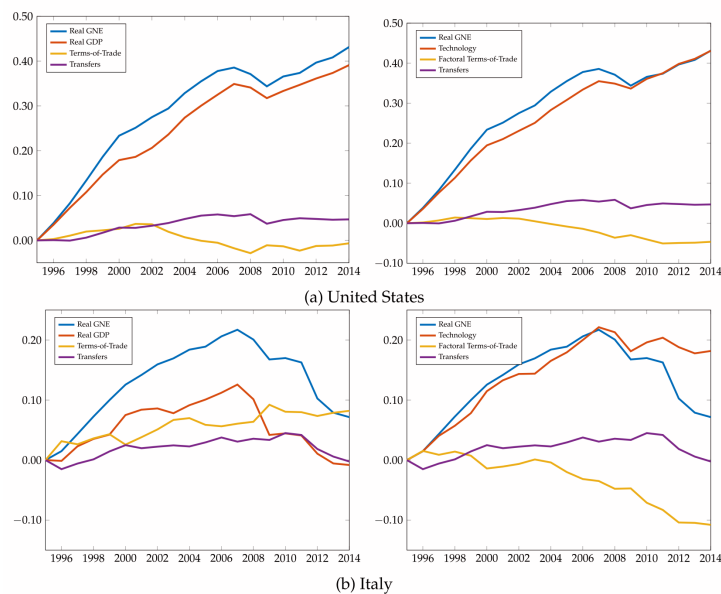


Figure 8: Original Figure 3: cumulative real GNE decompositions for the United States and Italy.

5.3 Figure 4: counterfactual losses from higher iceberg trade costs

Read:

- The benchmark is a calibrated world model with input–output linkages, complementarities, and factor mobility assumptions.
- Panel (a) removes input–output linkages; losses are much smaller than in the benchmark.
- Panel (b) removes complementarities; losses remain close to the benchmark, suggesting this dimension is less important in this calibration.
- Panel (c) removes factor mobility; many countries move above the 45-degree line, meaning losses are larger.
- Panel (d) imposes sticky wages; losses are far larger for many countries.

Economic message. The main quantitative message is that network structure and labor-market/nominal frictions can dramatically change the cost of deglobalization-type shocks.

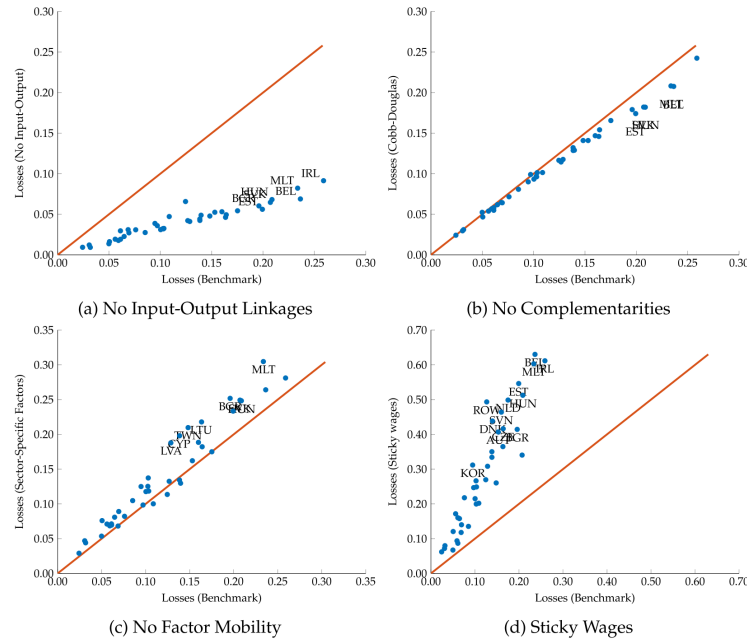


Figure 9: Original Figure 4: welfare losses from a 60 percent increase in iceberg trade costs under alternative assumptions.

6 Short summary

- Baqaee and Farhi develop a general framework for open-economy production networks with wedges.
- The framework covers tariffs, markups, iceberg costs, sticky prices, sticky wages, and other distortions.
- The main sufficient statistics are input–output shares, Domar weights, elasticities of substitution, factor income shares, and wedges.
- Real GDP and welfare are distinct objects in open economies.
- Real GDP reallocation effects vanish at efficient allocations, but welfare reallocation effects can remain because of factoral terms-of-trade.
- Theorem 3 generalizes hat algebra by characterizing first-order changes in every price and sales share.
- Iterating the first-order system gives an alternative way to compute exact nonlinear counterfactuals.
- Second-order tariff losses are Domar-weighted Harberger triangles.
- Input–output linkages, complementarities, sector-specific factors, and sticky wages can magnify the cost of trade shocks.
- Quantitative exercises show that removing input–output linkages understates losses and sticky wages can greatly magnify them.

7 Conclusion

7.1 Core conclusion

The paper shows that large-scale trade models do not have to be black boxes. Their responses to shocks can be decomposed into transparent sufficient statistics. Real GDP, welfare, prices, and quantities all move according to network exposure, substitution elasticities, wedges, and factor-income reallocations.

7.2 Economic intuition

A trade shock does not only affect the directly traded good. It moves through supply chains, changes input prices, changes sales shares, changes factor rewards, and reallocates income across households and countries. The aggregate effect depends on how important the shocked sectors are in the network and how easily the economy can substitute away from them.

7.3 Takeaway

The paper provides a practical bridge between simple theory and quantitative trade computation. The key lesson is that the cost of trade barriers depends not only on trade shares and elasticities but also on production networks, distortions, and redistribution through factor markets.